Class 7 SP250: Counting Part 2

Permutations: N! Is the number of ways to permute (or order) N objects.
Combinations: The number of ways to choose k objects out of N is "N choose k":

$$\binom{n}{k} = \frac{n\cdot(n-1)\cdot(n-2)\cdot \ldots \cdot(n-k+1)}{k!} = \frac{n!}{k!(n-k)!}$$

Problem 1: Sergey knows 7 combinatorics problems. How many ways are there to pick 3 of them and assign them to the students, if

(a) Order does not matter
(b) Order does matter – these are problems #1, #2, #3.

Answer: (a) $\binom{7}{3} = \frac{7\cdot6\cdot5}{3\cdot2\cdot1} = 35$

(b) $7 \times 6 \times 5 = 210$ or $\frac{7!}{(7-(7-3))!} = \frac{7!}{4!} = 7 \times 6 \times 5 = 210$

Problem 2: AMC 8, problem 16:
Professor Chang has nine different language books lined up on a bookshelf: two Arabic, three German, and four Spanish. How many ways are there to arrange the nine books on the shelf keeping the Arabic books together and keeping the Spanish books together?

Treat Arabic books as one group and Spanish books as one group. Then there are 5 objects to arrange: 3 German books, Arabic group and Spanish group.
There are 5! ways to arrange these 5 objects.
There are 2! ways to arrange the Arabic books in the Arabic group.
There are 4! ways to arrange the Spanish books in the Spanish group.
In total, there are 5! * 2! * 4! = 5,760 ways to arrange the books.

(Compare: Suppose there were no restrictions. Then it would be 9! = 362880 ways.)

Problem 3: There are 15 students in Math-m-addicts class.

a) How many ways are there to pick 5 of them? (answer form: "a choose b")
Answer: 15 choose 5.
b) How many ways are there to pick 5 of them, with one of them being Yirong?
Answer: "14 choose 4" ← need to pick 4 more out of 14.
c) How many ways are there to pick 5 of them and NOT pick Yirong?
Answer: "14 choose 5" ← need to pick 5 out of only 14.

Notice: $\binom{15}{5} = \binom{14}{4} + \binom{14}{5}$

→ Every committee of 5 people either contains Yirong or it does not. The LHS represents the total number of such committees and the RHS does as well but breaks it down by Yirong's participation.
There is nothing special about 15 and 5 in this example.

Pascal's Rule:

Theorem: For positive integers n, k with k ≤ n:

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

You must know this.

Problem 4: We have 5 people. How many ways are there to form a committee of some of them, with potentially no people or all 5?

Answer: 2^5 because each person has two choices: in or out.

How many ways are there to form a committee of k people? "5 choose k".

$$\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 2^5$$

Sum of the binomial coefficients:

$$\binom{n}{0} + \binom{n}{1} + \ \dots \ + \binom{n}{n-1} + \binom{n}{n} = 2^n$$

RHS represents how many distinct committees one can form with n people.
LHS breaks it down by size; the committee can have 0, 1, 2, … (n-1), or n people.

Symmetry rule:

$$\binom{n}{k} = \binom{n}{n-k}$$

Proof: choosing k people out of n is the same as NOT choosing (n-k) of them.

Counting and number theory

Counting divisors:

Problem: How many positive integer divisors does 120 have?

Answer: $120 = 2^3 * 3 * 5$.

- There are 4 choices for the power of 2: 0, 1, 2, 3.
- There are 2 choices for the power of 3.
- There are also 2 choices for the power of 5.

In total 120 has 4 * 2 * 2 = 16 positive integer divisors.

To find how many divisors N has: Factor N fully: **p's are primes and q's are their exponents:**

$$N = p_1^{q_1} \cdot p_2^{q_2} \cdot \ \dots \ \cdot p_k^{q_k}$$

(q_1+1) choices for power of p_1.
(q_2+1) choices for the power of p_2.
so on up to k.
Number of divisors = $(q_1 + 1) \cdot (q_2 + 1) \cdot \ \ldots \ \cdot (q_k + 1)$

Sketch of a proof: The idea is that every divisor of $N = {p_1}^{q_1} \cdot {p_2}^{q_2} \cdot \ \ldots \ \cdot {p_k}^{q_k}$ has the form $d = {p_1}^{m_1} \cdot \ldots \cdot {p_k}^{m_k}$ where $0 \leq m_j \leq q_j$, so where are $1 + q_j$ possibilities for those powers (it's one more than q_j because 0 is allowed).

Problem 1. How many divisors does 2^{10} have?
Answer: 11. There are 11 choices for the power of 2.
Problem 2. How many divisors does 4^8 have?
Answer: 17. 4 is not prime!
Problem 3. How many divisors does 60 have?
Answer: 12. $60 = 2^2*3*5$. (2+1)(1+1)(1+1) = 12.

Problem: (AMC 8, problem 17): How many positive divisors does 23,232 have?

Solution: 23,232 is clearly divisible by 2 and 3. Factor factor factor until you are stuck! $23232 = 2^6 * 3 * 11^2$. There are 7 * 2 * 3 = 42 divisors. There are 7 choices for the power of 2, 2 choices for the power of 3 and 3 choices for the power of 11.

Problem: There are 9 beach volleyball players. How many ways are there to pick 6 of them to play a 3 on 3 match?

Answer: There are "9 choose 3" ways to pick the first 3 players. There are "6 choose 3" ways to pick the second set of 3 players. The sets are interchangeable and so the answer is

$$\binom{9}{3} \cdot \binom{6}{3} \cdot \frac{1}{2} = 840$$

AMC 10, Problem 17: A child builds towers using identically shaped cubes of different colors. How many different towers with a height 6 can the child build with 2 red, 2 green and 3 blue cubes? (one cube will be left out).

- How many ways are there to build a 6-high tower with 2 red, 2 blue and 2 green cubes?
  $\frac{6!}{2!\cdot 2!\cdot 2!}$
  YOU NEVER MULTIPLY BEFORE DIVIDING. EVER.
  $\frac{6\cdot 5\cdot 4\cdot 3\cdot 2}{4\cdot 2} = 90$
- How many ways are there to build a 6-high tower with 1 red, 3 blue and 2 green cubes?
  $\frac{6!}{1!\cdot 2!\cdot 3!} = \frac{6\cdot 5\cdot 4\cdot 3\cdot 2}{2\cdot 3\cdot 2} = 60$
- Similarly, 60 when green is not used.

Total of 60 + 60 + 90 = 210.

Clever solution:

Stack up all 7 cubes. How many different towers?

$$\frac{7!}{2!\cdot 2!\cdot 3!} = \frac{7\cdot 6\cdot 5\cdot 4\cdot 3\cdot 2}{2\cdot 2\cdot 6} = 2\cdot 3\cdot 5\cdot 7 = 210$$

Why does this solution work? It works because two different arrangements of the 7 cubes lead to different arrangements of the top 6 cubes. If two different arrangements of 7 cubes had the top 6 cubes match, then the 7th cube has to match also since we are using 2,2,3 green, red and blue cubes.

Problem (AMC 10, problem 7): A student must choose a program of four courses from a menu of courses consisting of English, Algebra, Geometry, History, Art, and Latin. This program must contain English and at least one mathematics course. In how many ways can this program be chosen?

Solution: Use casework.

- E+A+G+(one other): $\binom{3}{1} =$ 3 choices.
- E+A+(two others, not geometry): $\binom{3}{2} =$ 3 choices
- E+G+(two others, not algebra): $\binom{3}{2} =$ 3 choices.

There are a total of **9 ways** to choose a program.